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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Examination July 2021 (2019 Scheme)



Course Code: EET206

Course Name: DIGITAL ELECTRONICS

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- | | | |
|----|--|-----|
| 1 | Convert | (3) |
| | a) $(7483)_{10}$ into hexadecimal | |
| | b) 1 1001 0100 into Gray Code | |
| 2 | Realize NOT, AND and OR gate using NAND gates only. | (3) |
| 3 | State and explain DeMorgan's theorem. | (3) |
| 4 | Prove that $A + A'B = A + B$. | (3) |
| 5 | Draw a 4-bit gray to binary code converter circuit. | (3) |
| 6 | Draw the block diagram of ALU. | (3) |
| 7 | Explain Preset and Clear inputs of a flip-flop. | (3) |
| 8 | Draw the logic diagram of a 4 bit Johnson counter and explain its working. | (3) |
| 9 | Draw and explain R-2R ladder type DAC | (3) |
| 10 | Differentiate between PLA and PAL | (3) |

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

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|----|---|-----|
| 11 | a) Subtract 46 from 99 using 1's complement and 2's complement methods. Compare both methods. | (7) |
| | b) Draw and Explain TTL NAND gate implementation | (7) |
| 12 | a) With neat diagram, explain the operation of CMOS NOR gate. | (7) |
| | b) Explain the error detection using parity method in digital transmission. Discuss how odd parity error detection carried out for transmitting the letter 'B' in ASCII code. | (7) |

Module -2

- 13 a) Design a full subtractor circuit using basic gates. (7)
 b) Reduce the expression using K map, (7)
 $F(A,B,C,D) = \sum m(6,7,8,10,11,15) + d(0,2,3,4,5,9,14).$
 14 a) Draw and explain four bit parallel adder/subtractor circuit. (6)
 b) Express $F(A,B,C,D) = \overline{A + B\overline{C} + C} + A\overline{C}$ in standard SOP and POS forms. (8)

Module -3

- 15 a) Realize a 2- bit comparator circuit. (8)
 b) Implement the function $F(A,B,C,D) = \sum m(0,2,4,7,9,14)$ using 4 X 1 MUX. (6)
 16 a) Differentiate decoder and encoder. Design a BCD to decimal decoder circuit. (8)
 b) Design an even parity generator circuit for 3-bit messages. (6)

Module -4

- 17 a) Design a mod-12 asynchronous counter using J-K flip flops. Draw the timing diagram. (8)
 b) Explain different types of shift registers. (6)
 18 a) Design mod-14 synchronous counter using T-flip flop by explaining the steps in detail. (10)
 b) Convert J-K flip-flop to T flip-flop. (4)

Module -5

- 19 a) Compare Mealy and Moore state machine models with example. (6)
 b) Explain the working of (8)
 (i) successive approximation ADC and
 (ii) Flash type ADC.
 20 a) Implement the function $F(A,B,C,D) = \sum m(3,7,8,9,11,15)$ using PLA (6)
 b) Implement AND gate and a half adder circuit using VHDL. (8)
